

CODINGS AT THE ORGANISM SIDE OF THE ORGANISM-ENVIRONMENT SYSTEM
Commentary on Jarvilehto on Efference-Knowledge

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ABSTRACT: Jarvilehto claims the empirical results he presents show that "the application of information theory to sensory function is problematic" and that knowledge of the environment does not come from sensory transduction alone. These are the right conclusions to draw from the experiments. But Jarvilehto does not relate them specifically to the idea of "neural codes," by which I assume he means something like internal mental representations. I spell out the consequences of Jarvilehto's conclusions for the idea of a neural code by way of friendly amendments to Jarvilehto's understanding of the organism-environment system.

KEYWORDS: afference, artificial life, efference, epistemology, evolution, Gibson, knowledge, motor theory, movement, perception, receptors, robotics, sensation, sensorimotor systems, situatedness

1. The consequences of Jarvilehto's empirical argument that "the application of information theory to sensory function is problematic" (21) are wide-ranging. One consequence he does not note explicitly is that if one cannot apply information theory to the senses, one also cannot apply it to the content of inner representational states. If inner representations are taken to have meaning in virtue of bearing information, such states must be reliably related to the world events about which they bear information (10): Dretske (1981), for example has argued that the probability of the world event given the internal state must be 1! Jarvilehto's experiments (along with those by Skarda and Freeman 1987) show that states are not reliably related to events in the world. This relegates an enormous amount of philosophical work on content to the scrap heap, including Dretske (1981), Markman and Dietrich (1998), and many, many others.

2. This does not mean that there is no sense in which certain of the internal states of the organism have content or are about that organism's environment. This conclusion, which Jarvilehto invites but does not explicitly draw and Clark (1998) rejects in his commentary, simply does not follow from Jarvilehto's experiments. For there are theories of content that do not tie representation constitutively to information from the senses. Such an understanding of representation would require that a state or feature of an organism meet three criteria to be a representation. First, it must be produced and used by specialized parts of the organism whose jobs are to produce and use such states for features. Second, it must be the function of the state to adapt the specialized consumer to some aspect of the organism's environment, normally by leading the organism to behave appropriately with this aspect of the environment. Third, there must be other such states, whose function it is to adapt the organism to other aspects of its environment; that is, the organism must be able to represent more than one aspect of its environment.

3. This understanding of representation is similar to Millikan's (1984) teleological theory of content (see also Millikan 1998). I will not defend this way of thinking of representation here, nor will I discuss the many attractive features that recommend it (I discuss this definition at length and defend it in Chemero 1998, Chapter 1). The one feature of the definition that is important for my purposes here is that it requires that we focus on the representation consumer in determining the content of a representation -- the content is the way the world would need to be for the behavior caused by the representation consumer to adapt the agent to the world (R2). Hence what some state of an organism represents may have nothing to do with what event in the world or at the organism's periphery causes it. Rather, it depends on the sort of behavior the state engenders.

4. Note that among the things that can meet the criteria of this theory of representation are what Clark (1997) calls "action-oriented representations" (AORs). These both describe a situation and suggest an appropriate reaction to it; as Clark puts it, they are maps that are also controllers (1997, p.49). AORs are in a certain sense more primitive than other representations in that they can lead to effective behavior without requiring separate representations of the state of the world and the cognitive system's goals. That is, the perceptual systems of agent need not build a detailed, action-neutral representation of the world, which can then be used by the action producing parts of the agent to guide behavior; instead, the agent produces representations that are geared toward the actions it performs from the beginning. Gibsonian affordances, Clark points out, are examples of AORs; a rabbit simply sees something like "opportunity-for-cover-over-there" behind a tree and need not represent each of "there is a tree," "one can hide behind trees," and "I want to hide now." It is likely that many non-human animals and some human sub-personal mechanisms use only action-oriented representations.

5. Now it is quite evident that both Jarvilehto's rabbits and his imagined organism do have entities in them that meet the criteria for representation as outlined above; they have action-oriented representations. This can be seen most clearly with the work by Alexandrov et al. (1986) on pedal-pressing rabbits, discussed in 17-21. The activity in their rabbit's optic nerve, which was present whether or not the rabbit was receiving visual input, is an action oriented representation of the afforded pedal presses. First, it has a normal producer: Alexandrov et al.'s surprising finding is that the cause of the activity is activation of ganglion cells, and not visual stimuli. The activity also has a normal user: the parts of the rabbit's brain that cause it to actually press the pedal. Second, the function of the ganglion's activity is to lead the rabbit to press the pedal. This is the case because Alexandrov et al. trained the rabbit to press the pedal, and such training would have led to changes in the rabbit's brain and optic nerve activity, causing the particular patterns involved in the rabbit's ability to press the pedal to be produced when a pedal-pressing opportunity arose. In other words, the particular patterns of optic nerve activity are part of the rabbit's repertoire because they tended to lead to pedal pressing, thus it is their function to do so. And third, the rabbit no doubt has many, many different patterns of optic nerve activity, which lead it to behaviors other than pedal pressing. So the optic nerve activity is a representation. Looking at the behavior it engenders should lead us to believe that it represents an afforded pedal press. (This argument went very fast. I make it more formally for Skarda and Freeman's 1987 model of rabbit olfaction in Chemero [1998], where I make parallel arguments regarding connectionist networks, situated robots, and other dynamical

models of cognition.)

6. So Jarvilehto has not shown that there are no representations or codings in the brain. What he has shown is that many such codings are not of action-neutral features of the environment. An organism's brain will likely contain only encodings of features of the environment that are relevant to its behavior, and it will only encode them as they relate to behavior, as "something to hide behind" rather than as "pine tree 37." He has also shown that such encodings cannot be constitutively tied to information from the environment, and hence that information-based understandings of representation and content are inadequate. These are both welcome findings.

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